

Serie 11

LINEAR MAPS AND MATRICES

1. Matrix multiplication. Consider the real vector space $M_{2 \times 2}(\mathbb{R})$.

(a) Compute the square of

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

(b) Find a formula for the entries of the n -th power of $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ for all $n \in \mathbb{N}$.

2. Linear maps as matrices. Let $\mathbb{R}[X]_n$ be the vectorspace of all polynomials of degree $\leq n$ with real coefficients.

1. Show that

$$F : \mathbb{R}[X]_n \rightarrow \mathbb{R}[X]_n, \quad p \mapsto p'' + p'$$

is a linear map, where p' denotes the derivative of p .

2. Determine the matrix of F with respect to the basis $(1, x, \dots, x^n)$ of $\mathbb{R}[X]_n$.

3. Matrices. Let K be a field.

(a) Consider the matrices

$$A = \begin{pmatrix} A_1 & 0_{k \times (n-k)} \\ 0_{(n-k) \times k} & A_2 \end{pmatrix} \in M_{n \times n}(K)$$

with $A_1 \in M_{k \times k}(K)$ and $A_2 \in M_{(n-k) \times (n-k)}(K)$ for some $k \geq 1$, and

$$B = \begin{pmatrix} B_1 & 0_{k \times (n-k)} \\ 0_{(n-k) \times k} & B_2 \end{pmatrix} \in M_{n \times n}(K)$$

with $B_1 \in M_{k \times k}(K)$ and $B_2 \in M_{(n-k) \times (n-k)}(K)$. Show that

$$A \cdot B = \begin{pmatrix} A_1 \cdot B_1 & 0_{k \times (n-k)} \\ 0_{(n-k) \times k} & A_2 \cdot B_2 \end{pmatrix}$$

(b) Let A be defined as above and assume that A_1 , respectively A_2 , is invertible as an element of $M_{k \times k}(K)$, respectively $M_{(n-k) \times (n-k)}(K)$. Show that A is invertible.

- (c) Consider the space U of upper triangular matrices in $M_{n \times n}(K)$. Show that the product of 2 elements of U is in U .

4. Hom as a vector space. Let V and W be finite-dimensional vector spaces over a field K .

- (a) Suppose that $2 \leq \dim V \leq \dim W$. Show that

$$\mathcal{S} := \{T \in \text{Hom}(V, W) \mid T \text{ is not injective}\}$$

is not a subspace of $\text{Hom}(V, W)$.

- (b) Suppose that $\dim V \geq \dim W \geq 2$. Show that

$$\mathcal{T} := \{T \in \text{Hom}(V, W) \mid T \text{ is not surjective}\}$$

is not a subspace of $\text{Hom}(V, W)$.

5. Change of bases. Consider the linear maps $\mathbb{R}^4 \xrightarrow{f} \mathbb{R}^2 \xrightarrow{g} \mathbb{R}^3$ given by

$$f : \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \mapsto \begin{pmatrix} x_1 + 2x_2 + x_3 \\ x_1 - x_4 \end{pmatrix} \quad \text{and} \quad g : \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mapsto \begin{pmatrix} x_1 + x_2 \\ x_1 - x_2 \\ 3x_1 \end{pmatrix}.$$

Let

$$\mathcal{A} := \left(\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \\ 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 3 \\ 0 \end{pmatrix} \right),$$

let $\mathcal{B}$ be the standard basis of $\mathbb{R}^2$, and let

$$\mathcal{C} := \left(\begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \right).$$

1. Show that $\mathcal{A}$ is a basis of $\mathbb{R}^4$ and that $\mathcal{C}$ is a basis of $\mathbb{R}^3$.
2. Determine $g \circ f$ and the matrices
 - (i) of f with respect to the bases $\mathcal{A}, \mathcal{B}$.
 - (ii) of g with respect to the bases $\mathcal{B}, \mathcal{C}$.
 - (iii) of $g \circ f$ with respect to the bases $\mathcal{A}, \mathcal{C}$.

6. Linear maps. Let V be a vector space over a field K . Suppose that T_1 and T_2 are two linear maps from V to K that have the same kernel. Show that there exists a constant $c \in K$ such that $T_1 = cT_2$.

Hint: Use Serie 10 exercise 1.

7. Linear maps as matrices. Denote $\mathbb{R}[x]_d$ the set of polynomials over $\mathbb{R}$ of degree lower or equal to d . Suppose that $D \in \text{Hom}_{\mathbb{R}}(\mathbb{R}[x]_3, \mathbb{R}[x]_2)$ is the differentiation map $Dp = p'$. Find a basis of $\mathbb{R}[x]_3$ and a basis of $\mathbb{R}[x]_2$ such that the matrix of D with respect to these bases is

$$\begin{pmatrix} 0 & 1 & -1 & -1 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

Multiple Choice Questions. More than one answer can be correct.

Question 1. Let $\mathcal{A}$ and $\mathcal{B}$ be bases of $\mathbb{R}^2$ and denote $\{e_1, e_2\}$ the standard basis. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear map given by the matrix

$$M = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

with respect to $\mathcal{A}$ as a basis of the domain and $\mathcal{B}$ as a basis of the codomain. Which of the following statements are true?

- If $\mathcal{A} = \mathcal{B} = \{e_1, e_2\}$, f is a rotation around the origin.
- If $\mathcal{A}$ is the standard basis and $\mathcal{B} = \{e_2, -e_1\}$, f is a symmetry with respect to the point $(0, 0)$.
- If $\mathcal{A}$ is the standard basis and f is the identity, then $\mathcal{B} = \{-e_2, e_1\}$.
- If $\mathcal{B}$ is the standard basis and f is the symmetry with respect to the y -axis, then $\mathcal{A} = \{e_2, e_1\}$.

Question 2. Which of the following statements are true?

- Let V be an n -dimensional vector space over $\mathbb{R}$. The map $V \rightarrow \mathbb{R}^n$, $v \mapsto [v]_{\mathcal{B}}$, that sends any vector $v \in V$ to its coordinate vector with respect to a basis $\mathcal{B}$ of $\mathbb{R}^n$ is linear.
- Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be linear with $\text{Ker}(f) \neq \{0\}$ and $\text{Im}(f) \neq \{0\}$. Then there exists a non-trivial vector v that is in the kernel and in the image of f .
- If the kernel of a linear map $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is trivial, the map is invertible.